Practical Solutions to Protect Data Privacy

Data Saturday Columbus

August 16, 2025

We’ll cover:

Defining Data Privacy

- Why it’s Important

Disclosure Risk

Privacy-enhancing techniques

- Masking
- Differential Privacy
- Text redaction and replacement
- Synthetic Data

Next Steps

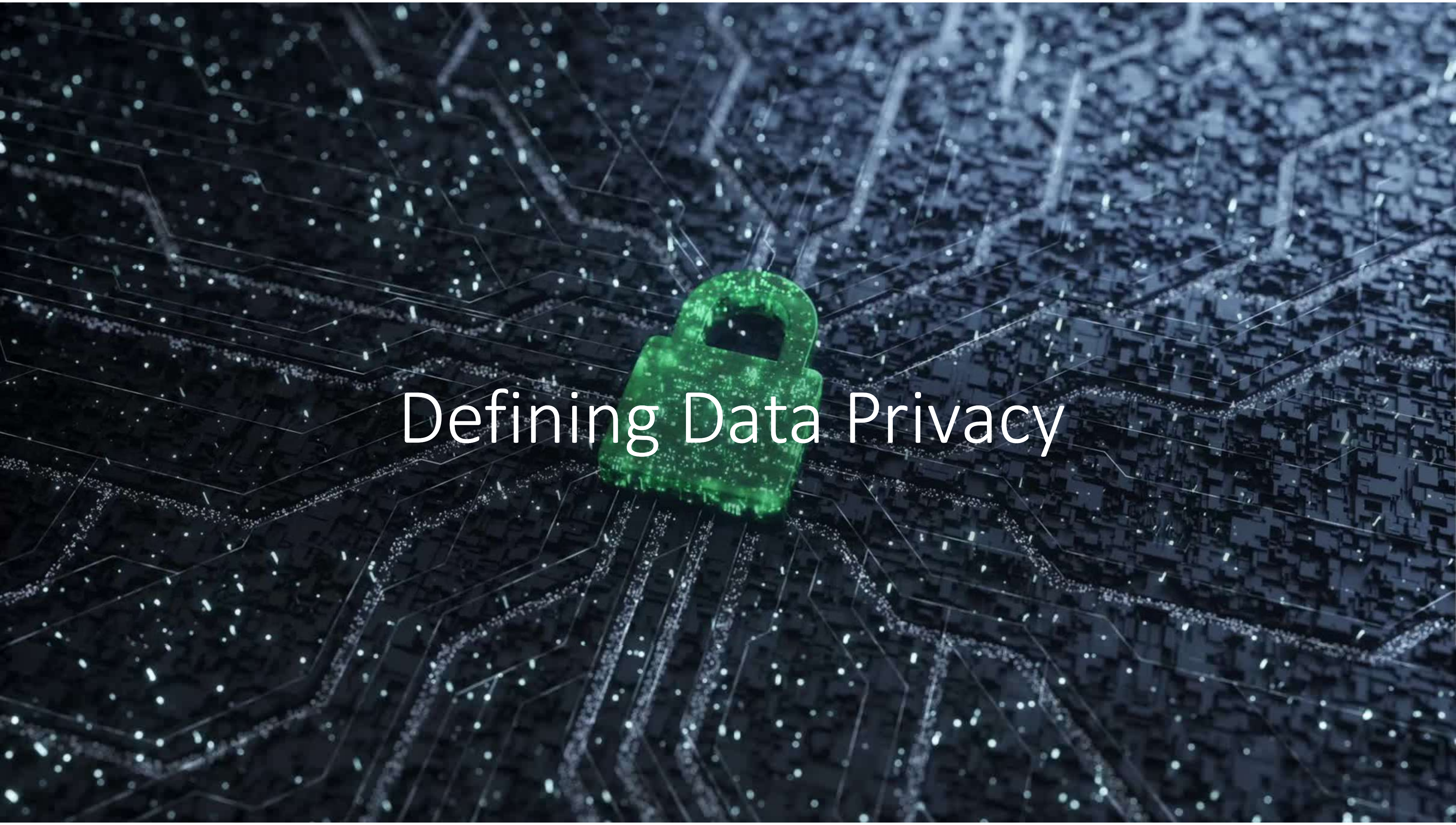
Hi!

Britton Gray

- Data professional for almost 20 years
  - My passions
    - Advanced analytics (data and language)
    - Data privacy
    - Data engineering

I’m an <u>actual</u> scrum master!

Defining Data Privacy



Data Privacy (*n*):

Any information about an **identified or <u>identifiable</u>** natural person

Data privacy rights:

Who has access to their information

What information can be accessed

When: Data kept only as long as it's legitimately needed

Where: Limitations on sharing and where data is stored

Why: the purpose(s) for storing/accessing that information

Legitimate Purposes

Valid reasons for storing and processing data (GDPR)

✅	**Affirmative consent of the data subject**
❤️	Vital interests of a person
📜	Fulfillment of a contract
⚖️	Legal obligation
🌎	In the public interest
🏢 ⇔ 🧍	Legitimate interests of processor – balanced by rights of the data subject

Why is this important?

Ethics!

Legal sanctions

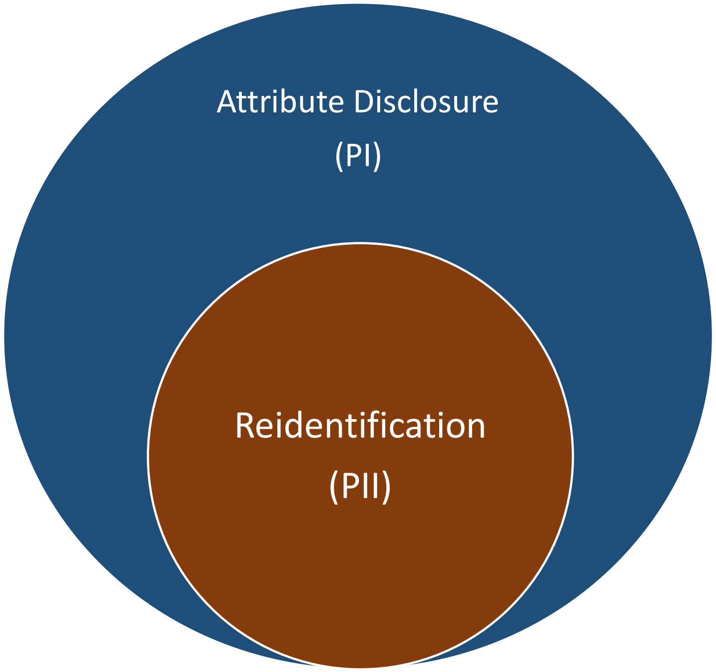
- Fines
  - \$2500-\$7500 per person affected under some state laws
  - GDPR: Max €20 million / 4% of annual turnover
- Consent decrees
- **Private lawsuits**


Disclosure Risk

Disclosure Types

<u>Reidentification</u>
A specific person is identifiable

<u>Attribute Disclosure</u>
One or more sensitive attributes or behaviors about a person become discoverable



"100% of our students met standards on the state exam!"

- Actual school who sued the state because they couldn't report it

“Anonymized” Medical Record

All identifiers have been removed

Hospital	162: Sacred Heart Medical Center in Providence
Admit Type	1: Emergency
Type of Stay	1: Inpatient
Length of Stay	6 days
Discharge Date	Oct-2011
Discharge Status	6: Dsch/Trfn to home under the care of a health service organization
Charges	$71,708.47
Payers	1: Medicare
	6: Commercial insurance
	625: Other government-sponsored payers
Emergency Codes	E8162: motor vehicle traffic accident due to loss of control; loss control mv-mocycl

Diagnosis Codes	80843: closed fracture of other specified part of pelvis
	51851: pulmonary insufficiency following trauma & surgery
	86500: injury to spleen without mention of open wound into cavity
	80705: closed fracture of rib(s); fracture five ribs-close
	5849: acute renal failure; unspecified
	8052: closed fracture of dorsal [thoracic] vertebra without mention of spinal cord injury
	2761: hyposmolality &/or hyponatremia
	78057: tachycardia
	2851: acute posthemorrhagic anemia

Age in Years	60
Age in Months	725
Gender	Male
ZIP	98851
State Reside	WA
Race/Ethnicity	White, Non-Hispanic
Procedure Codes	5781: Suture bladder laceration
	7939: 7919: Open/Closed reduction of fracture of other specified bone
Physicians	...
...	...

Newspaper Article

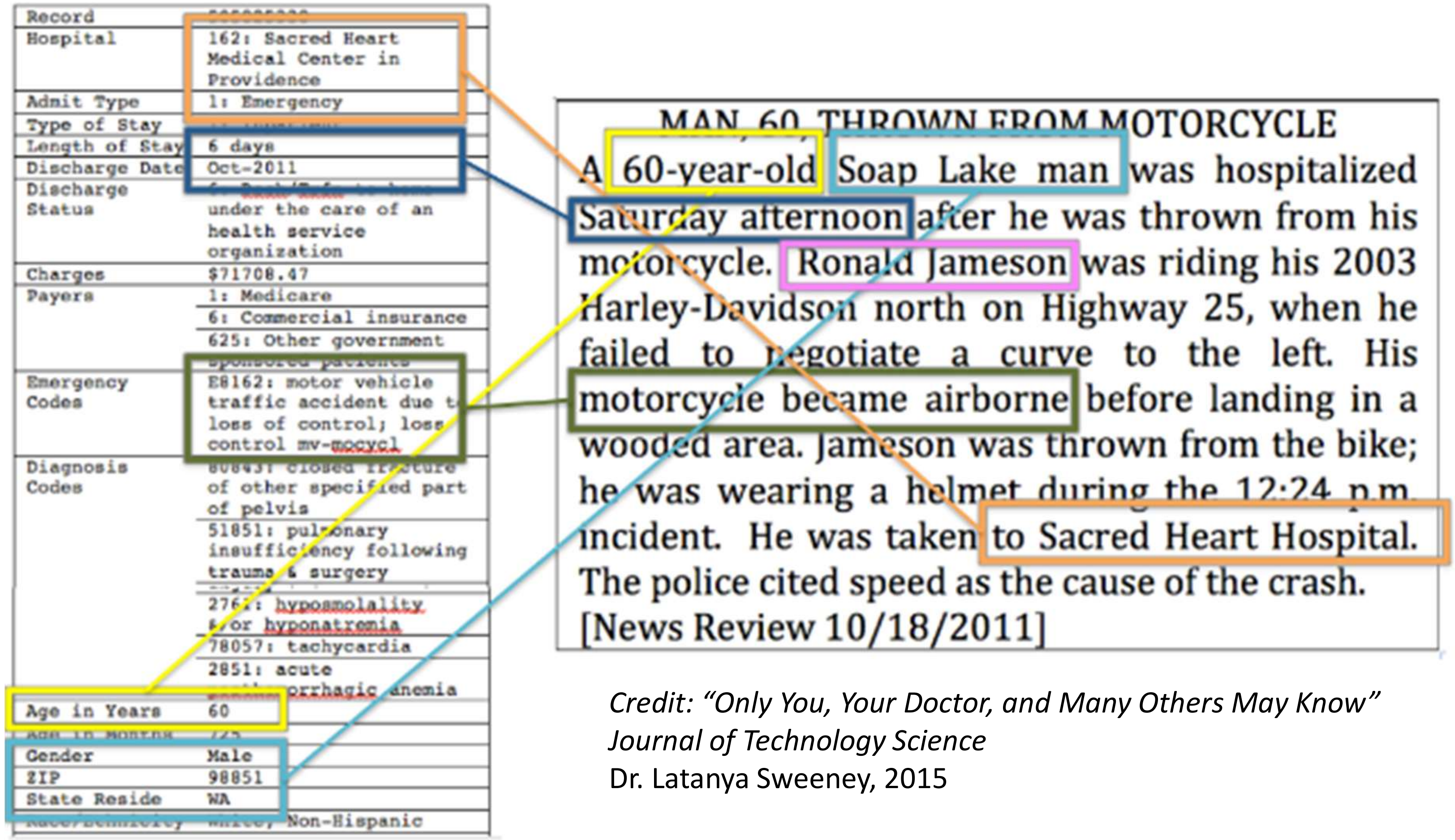
MAN, 60, THROWN FROM MOTORCYCLE

A 60-year-old Soap Lake man was hospitalized Saturday afternoon after he was thrown from his motorcycle. Ronald Jameson was riding his 2003 Harley-Davidson north on Highway 25, when he failed to negotiate a curve to the left. His motorcycle became airborne before landing in a wooded area. Jameson was thrown from the bike; he was wearing a helmet during the 12:24 p.m. incident. He was taken to Sacred Heart Hospital. The police cited speed as the cause of the crash.

[News Review 10/18/2011]

(name changed)

Reidentification → Attribute Disclosure



Record	[illegible]
Hospital	162: Sacred Heart Medical Center in Providence
Admit Type	1: Emergency
Type of Stay	[illegible]
Length of Stay	6 days
Discharge Date	Oct-2011
Discharge Status	[illegible] under the care of an health service organization
Charges	$71708.47
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	6: Commercial insurance
	625: Other government sponsored patients
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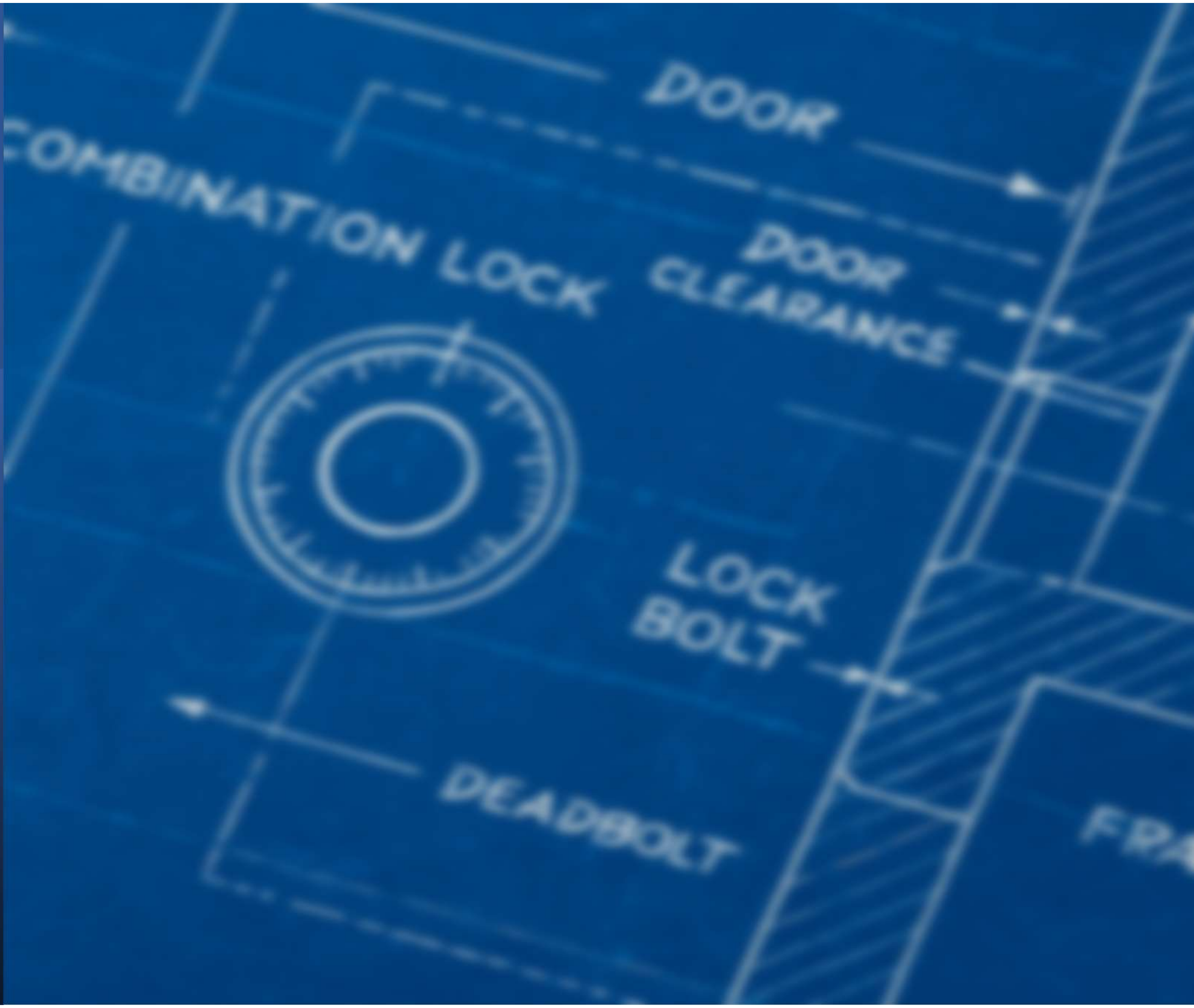
Credit: "Only You, Your Doctor, and Many Others May Know"
Journal of Technology Science
Dr. Latanya Sweeney, 2015

Techniques



Masking


Email
State
b***@g***.com
Indiana
j****@y***.com
California
m***@o***.com
Texas
s***@g
New York
p***@y***.com
Florida
a***@o***.com
Michigan
k***@y***.com
Illinois
t***@g***.com
Pennsylvania

Masking

- Useful for individual records
  - Be aware of some aggregation issues (GROUP BY)
- Works with column / user / row-level security
- Two types of masking targets:
  - Would-be or could-be identifiers
  - Sensitive attributes
- Efficacy heavily depends on what and how you mask

Masking Methods

Mask	Example	Usage Notes
Redaction	Britton → NULL	No need to do any analysis on the field, NULL may convey mistaken meaning.
Constant Masking	Britton → ****	Same as above, but mask is obvious
Partial Masking	bgray@pltw.org → b***@p***.org 8/6/2024 → 1/1/2024 12.34.56.78 → 12.34.56.xx	Decreases reidentification risk, but doesn't eliminate it Useful when some aggregation needed
Hashing	Britton → 1e00fea58e20cc1	**Works with aggregates.** Sliceable data that's obviously masked. Salt the hash: ("AA1438-Britton")
Banding / Blurring	33 → 25-34 (discrete) 33 → 30 (aggregable)	Numeric variables important to analysis but pose reidentification risk
Geo Blurring	~~39.9155°N, 86.0650°W~~ 39.92°N, 86.07°W	**Treat precise geo data as identifying.** 1/100 of a degree = about ½ mile

Masking in Data Platforms

Implemented via **masking policies**

- Define once, use multiple times
- Anything you can do in a CASE statement

Masking functions written in SQL

- Applied as a part of column definition

Configurable pre-defined masking functions, applied to columns

- Implemented in Fabric's SQL analytics endpoint and Warehouse

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Not native; ideally use OLS / RLS

Idea: Source view, UNION masked and unmasked, use RLS DAX

Same as Power BI

- Understand who can see what

Suppression

- Masking's cousin
- Works on aggregated data
- Guards against attribute disclosure

The problem:

Everyone in OH makes more than $50K!

	< $50K	$50-100K	$100K+	Total
Ohio	0	12	14	**26**
Indiana	6	14	2	**27**
Total	**6**	**26**	**21**	**<u>53</u>**

Suppression Guidelines

Step One:

- Suppress any aggregates if any population size (N) of less than 5-10

	< $50K	$50-100K	$100K+	Total
Ohio	*	12	14	**26**
Indiana	6	14	*	**27**
Total	**6**	**26**	**21**	**53**

Suppression Guidelines

Step Two:

- If you have <u>any</u> small values*:*
  - Keep suppressing until your total population is greater than threshold
  - Then suppress the next lowest value
- Avoid using subtotals

	< $50K	$50-100K	$100K+	Total
Ohio	*	12	14	*
Indiana	*	14	*	*
Total	*	**26**	*	<u>**53**</u>

Suppression: Data Platforms

Aggregation policies

- Disallows non-aggregate queries on a table
- Suppresses any aggregated results with fewer than *n* rows

Nothing built-in

Pre-aggregate results then apply masking logic

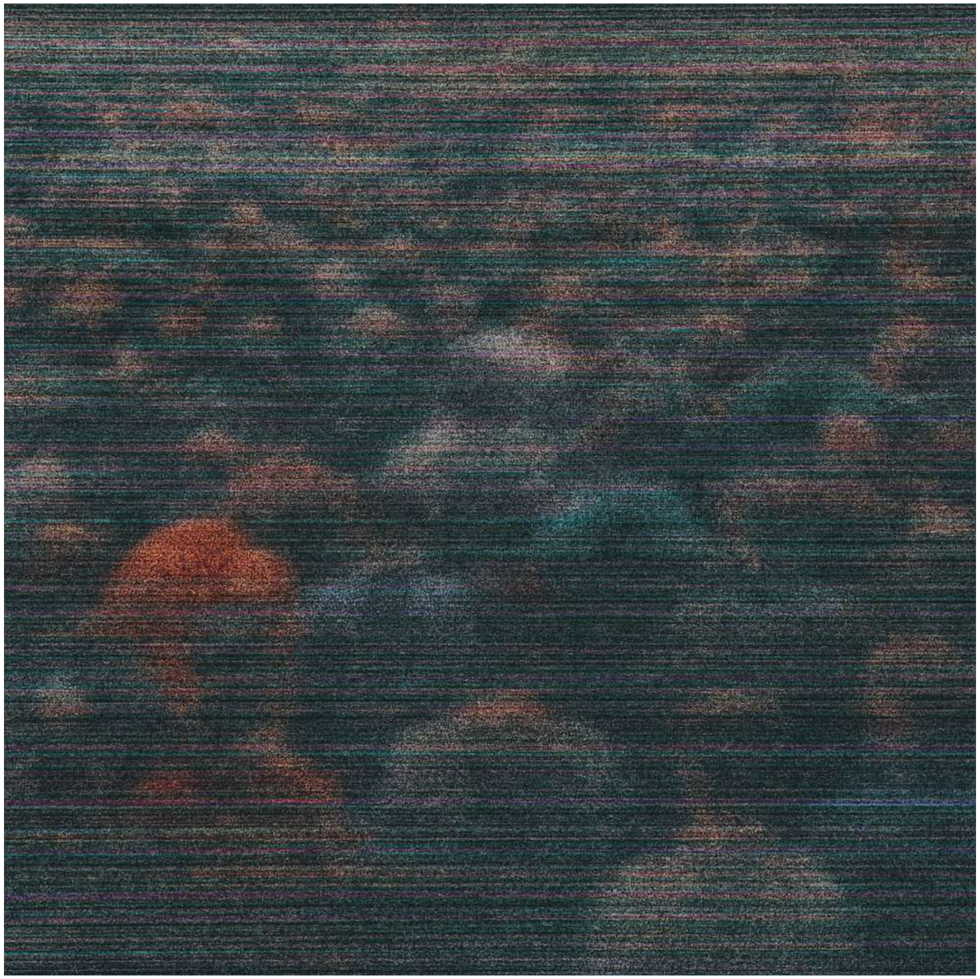
Use Python visual

Possible via TabPy

- Pass the data set in, get a suppressed one back

Differential Privacy

Adding noise



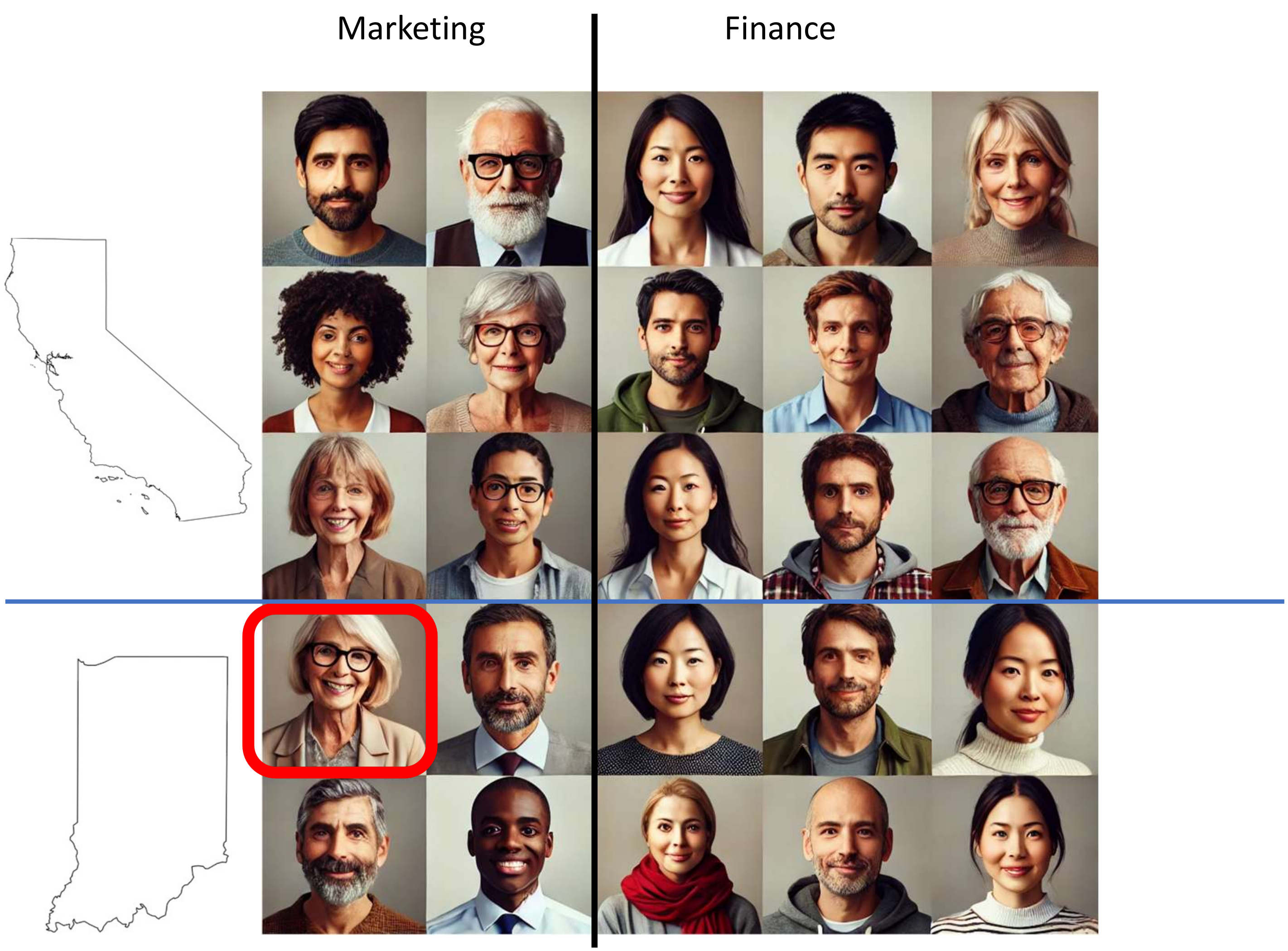
A quick game

<u>Find the salary:</u>

```
SELECT
   AVG(SALARY),
   COUNT(*)
FROM EMPLOYEES
WHERE . . .
```

Do so with….

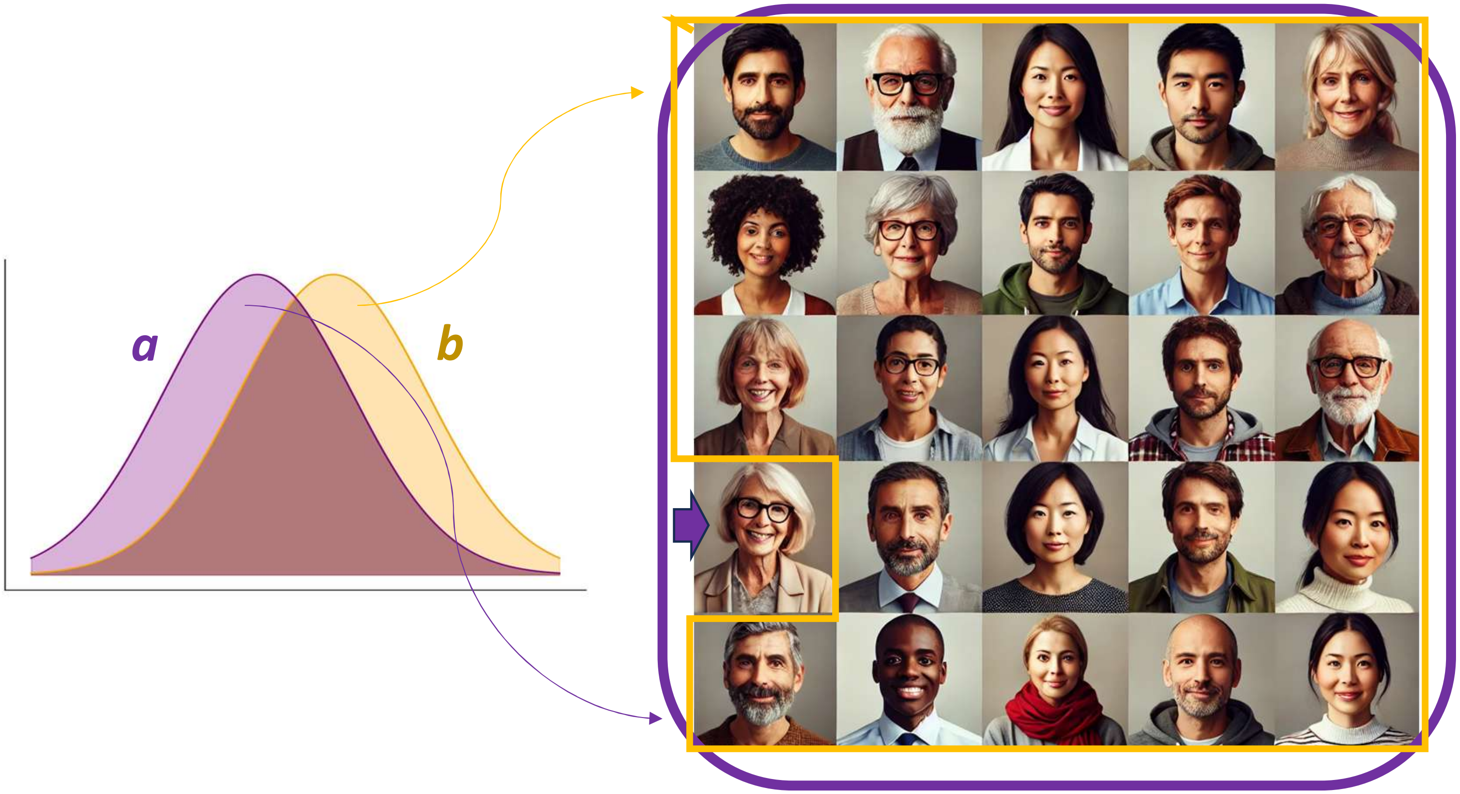
2 Queries, return 5+ rows



(specifically, a **differencing attack**)

Differential Privacy: Definition

- Adding statistical perturbation (noise) to aggregated query results



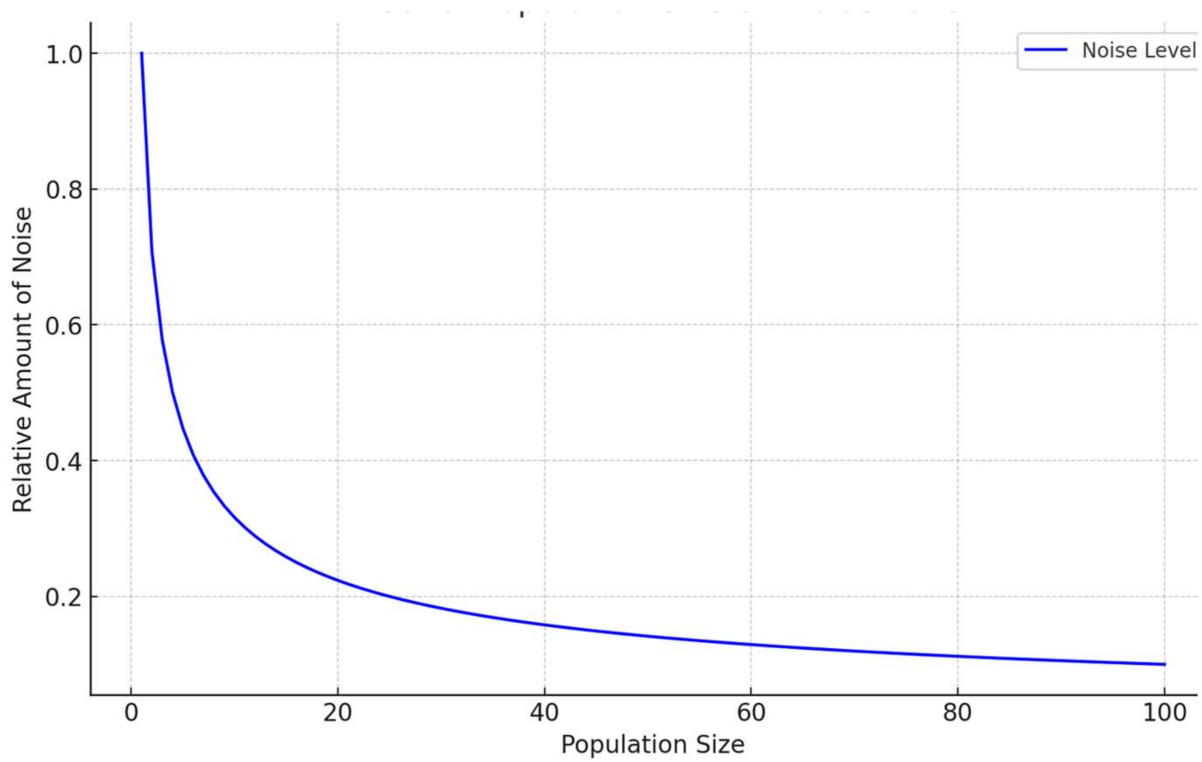
a
is **indistinguishable**
from ***b***

The What & Why Differential Privacy

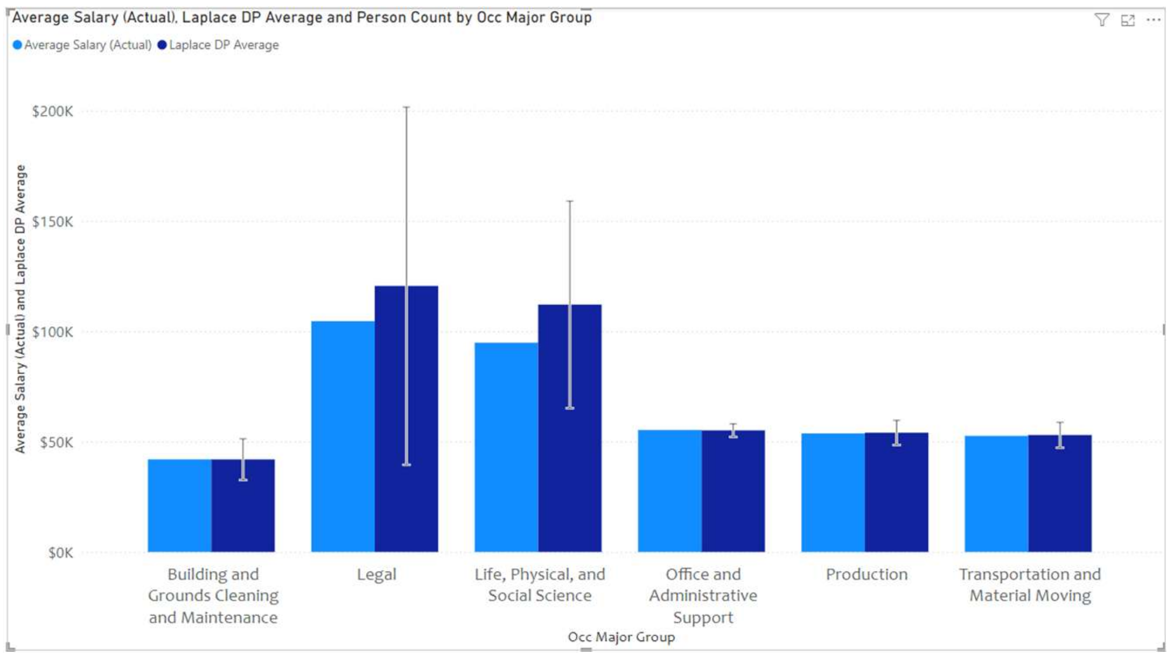
- Used in aggregate queries only
- Guards against differencing attacks
- It’s still possible to uncover protected attributes through brute force
  - Countermeasure: **Privacy Budgets**

When Using Differential Privacy

- Phrase results as an interval
- Average salary: $90,538 ($88,517 - $92,559)
- More people = narrower range
- True differential privacy will infuse some “empty cells” with data



Diff Privacy: Example



Job Category	# People	Estimate Width
Legal	8	$162K
Maintenance	51	$19K
Office & Admin Support	240	$6K

Differential Privacy: Data Platforms

Snowflake: implemented as a privacy policy

Designed with active privacy attacks in mind, so it's rigid

Currently has several limitations; makes machine SQL difficult

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Notebooks and clean rooms – third-party / open-source libraries

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May be third party solutions

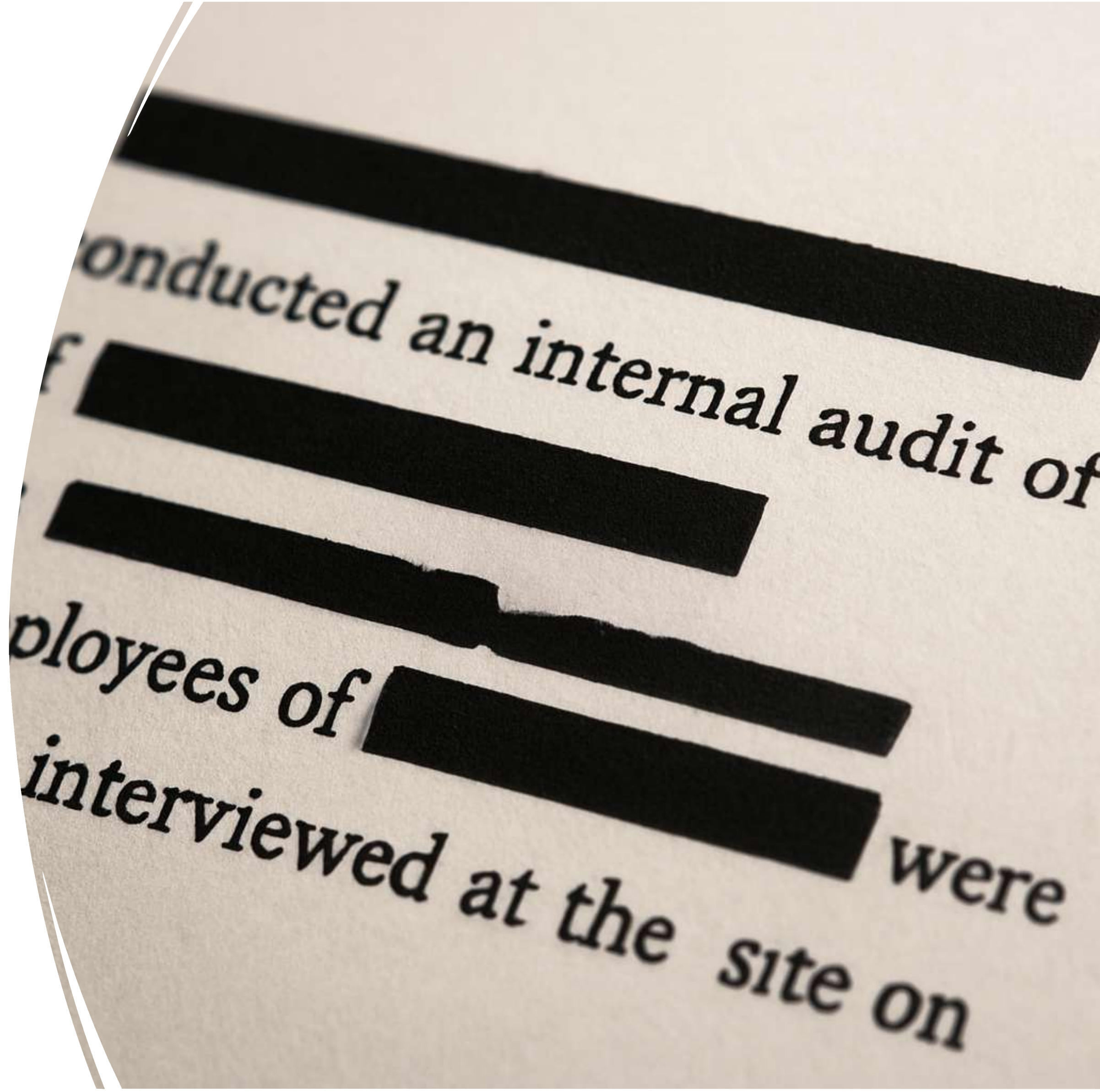
- Can use in-database Python with PyDP library

Can be approximated in both technologies

- More useful against accidental disclosure
- Will not be safe from an active privacy attack

Redaction

Preserving privacy in qualitative data



Qualitative Data

- **Our native language**: stories, conversations, context
- Explains the *why* behind numbers and trends
- Found in surveys, transcripts, documents, and messages
- Enables questions like: *“What did our detractors say?”*
- **AI unlocks analysis at scale**

Automated Redaction

- Uses language models to find PII
  - **Named Entity Recognition** (NER)
  - Small Language Models that look at words **in context**
  - Fights entity and attribute disclosure

organization

August attended a data conference at Otterbein University in August.

person

NOT a person

[Demo time!]

Common Tools

- Implemented via **Small Language Models** (SLMs)
- BERT is a family tree
  - DistilBERT-NER
  - RoBERTa
  - and many more
- All support fine-tuning
- Generally, > 90% accuracy

Azure AI Foundry

Google
BERT

spaCy

Recommendations

- Integrate redaction into data pipeline
- If you DIY, chain models for maximum efficacy
  - Written text: “cased” and “uncased”

Implicit Identifiers

"A data privacy speaker ate four tubs of Rocky Road ice cream last week after refereeing a rugby match."

Replacing Implicit Identifiers

Original Sentence:

"A data privacy speaker ate four tubs of Rocky Road ice cream last week after refereeing a rugby match."

Generalize:

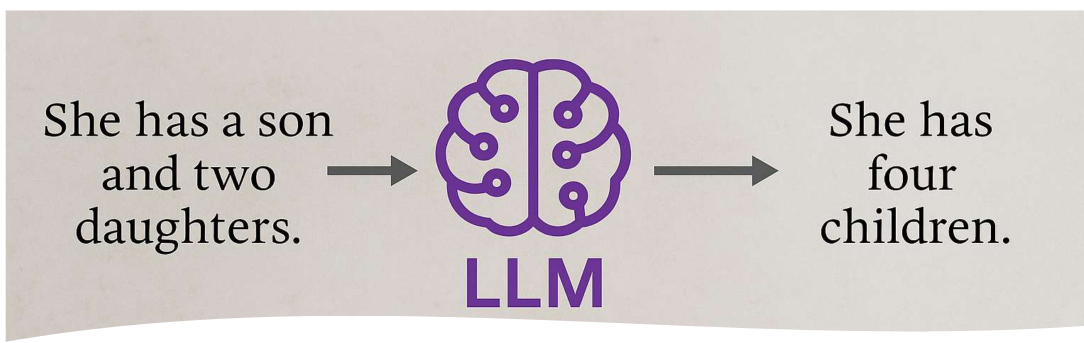
"A technology expert indulged in several desserts last week after participating in a sporting event."

Modify:

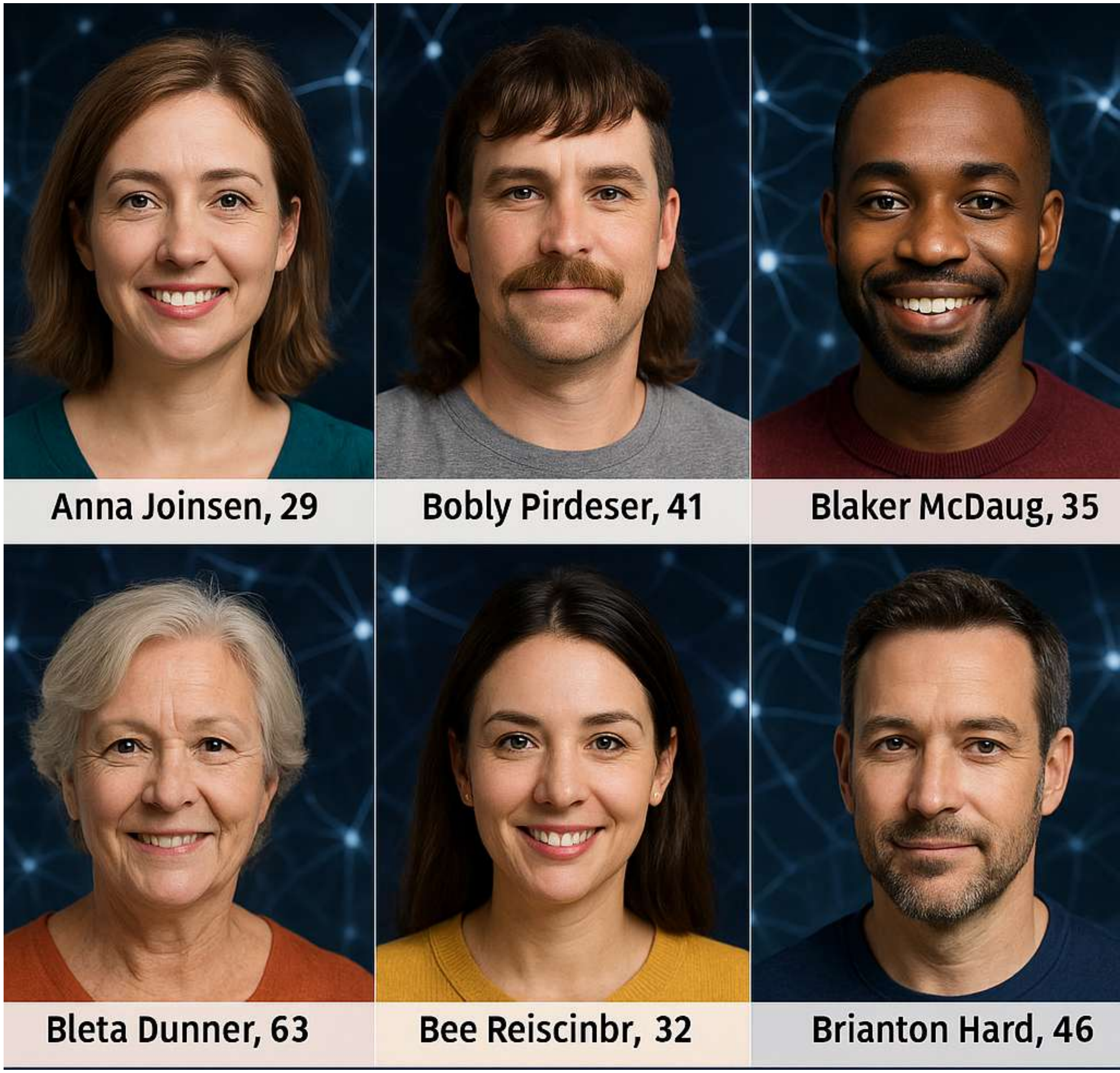
"A cybersecurity analyst ate five cartons of frozen yogurt last month after officiating a soccer game."

Using LLMs to Replace Implicit Identifiers

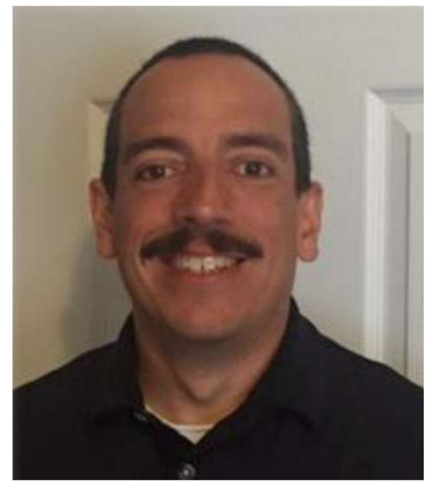
- **Can protect against both entity and attribute disclosure**
- LLMs can process text to identify and modify identifiers
  - Tend to replace aggressively
- Tip: output sanitized text in structured formats like JSON.
  - Mitigates the effect of including reasoning in the response
- Models constantly improving, but can be expensive to run
  - Some models can be run locally, e.g., Mistral 7b and gpt-oss
  - Running locally may be a plus, as data never leaves your environment



Synthetic Data

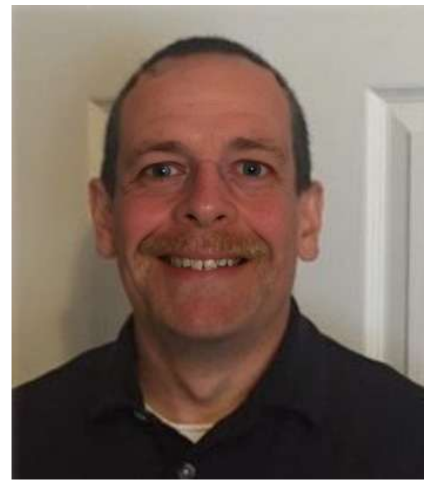


Britton



Warren

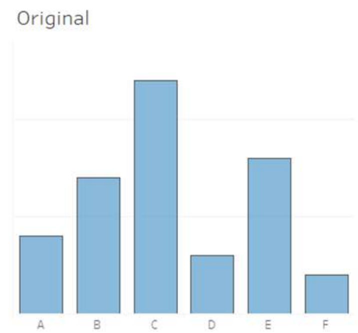
Britren

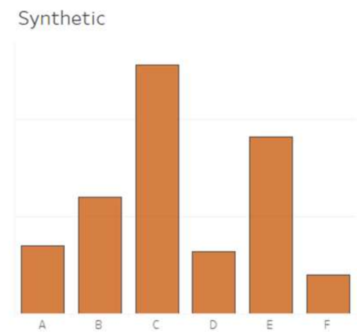


Warton

What is it?

- A new data set generated using real data
- Statistically consistent with actual data
- **Retains no actual source data** (or some if desired)
- It can miss rare patterns in data
- No entity or attribute disclosure
- Analyze synthetic data exactly as you would use original data

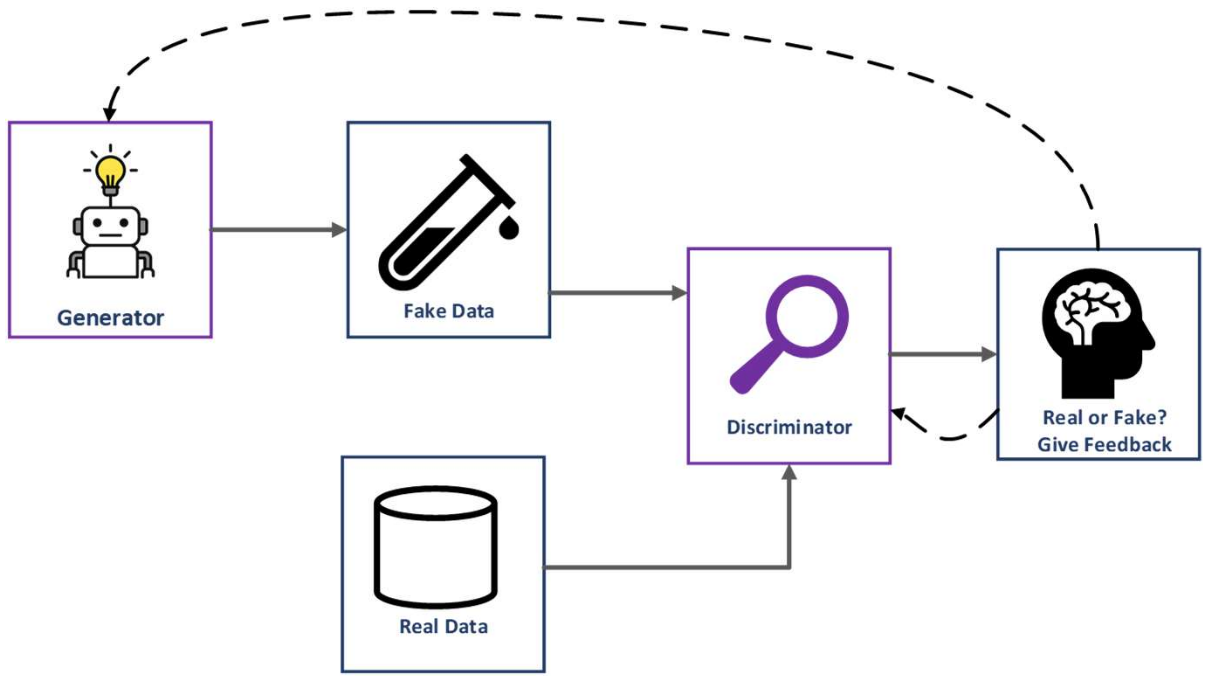




How is it created?

Most modern tools use **Generative Adversarial Networks (GANs)**

The generator iteratively improves fake data to fool the discriminator trained on real data.



Tools and Platforms

- **Snowflake**: GENERATE_SYNTHETIC_DATA
- **Paid tools** *(some offer free limited-user plans)*

  gretel hazy MDCLONE MOSTLY·AI

- **Open-source tools**

  gretel YData CTGAN datacebo SDV datacebo The Synthetic Data Vault

- Some require flat data sets, some allow tabular sets

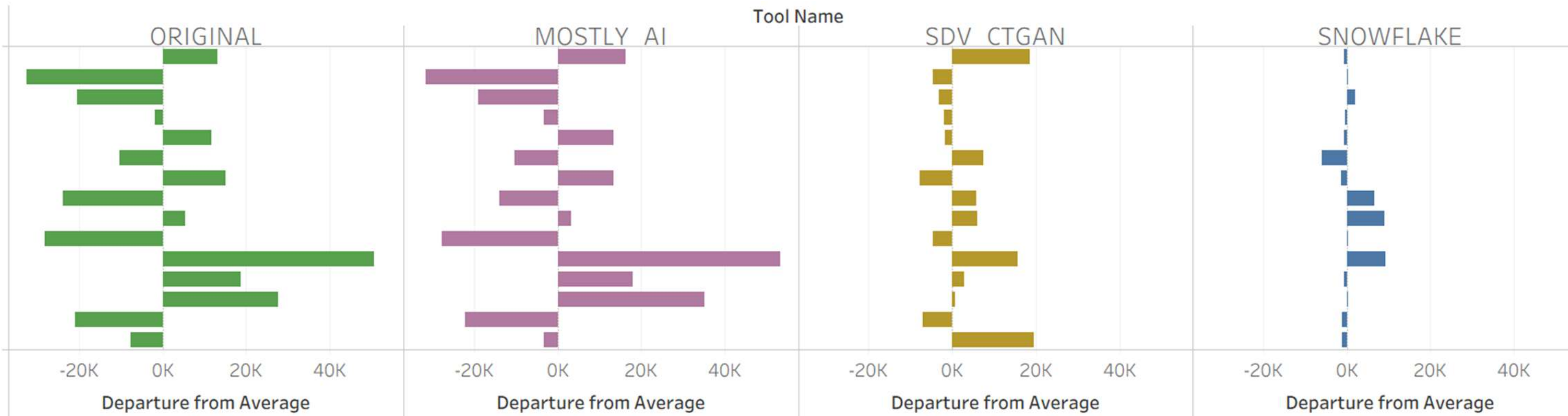
Example

Table columns

Include	Name	Encoding type
on	dim_person_key	Primary key
on	first_name	Language/Text
on	last_name	Language/Text
on	type	Tabular/Categorical
on	gender	Tabular/Categorical
on	approx_age	Tabular/Numeric: Auto
on	pop_score	Tabular/Numeric: Auto
on	address	Language/Text
on	city	Language/Categorical
on	state	Tabular/Categorical
on	zip	Language/Text
off	latitude	Tabular/Numeric: Auto
off	longitude	Tabular/Numeric: Auto
on	dim_occupation_key	Tabular/Categorical
on	final_salary	Tabular/Numeric: Auto

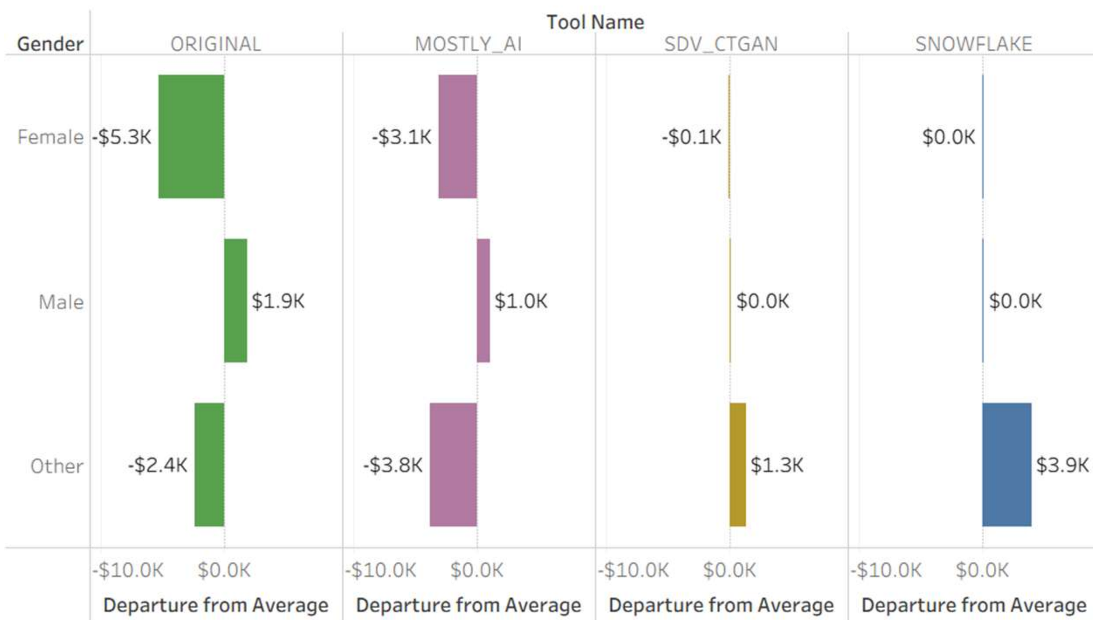
Average “Salary” Comparisons

By Occupation



Average “Salary” Comparisons

By Gender



Key Insights

Summary of Methods

Tehnique	Best For	Limitation
Masking	Structured, low-complexity data	Can be reversible
Differential Privacy	Aggregate queries	Can thwart via brute force; needs privacy budgets
Redaction (NER)	Transcripts, comments, and other unstructured text that may contain PII	Not 100% accurate, resource-intensive
Replacement	Unstructured text that may contain implicit identifiers	LLMs can get aggressive and resource-intensive
Synthetic Data	Enabling model training and analytics without using real data.	Outliers not well-represented

Next Steps

- Assess your data for privacy concerns
- Choose methods based on data type and use case
- Pilot modern approaches to unlock more value
- Build privacy into your data lifecycle: “Privacy by Design”

Thank you!

Questions?

GitHub Repo: https://github.com/IDreamInSQL/data_privacy_resources

brittongray